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A Phase 1, First-in-Human, Combined IND/IDE Clinical Trial Protocol

On-Premises LLM-Directed Robotic Pancreaticoduodenectomy (Whipple) with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma

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Protocol version: v4.0.0 (DRAFT scaffold) **Date:** June 20, 2026 **Regulatory basis:** IND (21 CFR part 312) + IDE (21 CFR part 812), Physical AI Subpart J

Independent research draft. Not an active IND or IDE, not an approved protocol, and not medical or regulatory advice; not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. The DOI placeholder 10.5281/zenodo.xxxxxxxx is filled at deposit. Clinical figures derive from the author’s simulation sources and are illustrative unless tied to a cited reference.

Keywords pancreatic ductal adenocarcinoma; robotic Whipple; daraxonrasib; Physical AI; on-premises LLM; IND/IDE; Phase 1; VVUQ

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1 Statement of Compliance

This trial will be conducted in accordance with the protocol, the principles of the International Council for Harmonisation Good Clinical Practice (ICH GCP), and applicable United States regulations governing both the investigational drug and the investigational device, together with the Physical AI overlay.

[DRAFTING INSTRUCTION] Carry the dual-citation compliance statement. Drug arm: 21 CFR parts 50, 54, 56, and 312 (IND). Device arm: 21 CFR parts 11, 50, 56, 812 (IDE, significant-risk per 21 CFR 812.3(m)). Physical AI overlay: 21 CFR part 312 Subpart J (sections 312.400-312.405) from *inputs/21cfr312_adapt/05_clinical_holds_appendices_closing.tex*. Add the ICH E6(R3) GCP attestation and the 21 CFR part 11 electronic-records clause supporting the hash-chained audit trail.

[DRAFTING INSTRUCTION] State that the study is registered as Interventional, Phase: Not Applicable for the device feasibility component while the drug component is Phase 1, Primary Purpose Device Feasibility plus dose-finding, per *research/ChatGPT-5.5-Thinking-Extended-19Jun26.md*. Include the signature blocks for the Sponsor-Investigator and the site Principal Investigator.

[DRAFTING INSTRUCTION] Render *mermaid/fig-03-ind-ide-pathway.md* here as a TikZ *mermaidfig* to show the combined IND/IDE compliance spine with the Physical AI Subpart J overlay.

2 Protocol Summary

2.1 Synopsis

A first-in-human, open-label, single-arm, dose-finding study of perioperative daraxonrasib combined with on-premises LLM-directed eight-arm robotic pancreaticoduodenectomy in adults with resectable or borderline-resectable KRAS-mutated PDAC.

[DRAFTING INSTRUCTION] Fill the Synopsis as a compressed mirror of the whole protocol: title, brief rationale, objectives/endpoints (from *sec-03*), design (single-arm, 3+3, up to n=18; from *sec-04* and *sec-09*), population (from *sec-05*), intervention (from *sec-06*), and duration (24-month OS follow-up). Pull headline numbers from *inputs/2030-pdac-1min-final-paper/sections/{introduction,results}.tex*.

2.2 Schema

[DRAFTING INSTRUCTION] Render *mermaid/fig-01-trial-schema.md* as a TikZ *mermaidfig* (screening -> consent with Physical AI opt-out -> Phase 0 simulation validation -> dose-cohort -> robotic Whipple -> perioperative advisory -> follow-up -> end of study).

2.3 Schedule of Activities (SoA)

[DRAFTING INSTRUCTION] Build the SoA as a full-width table using `>\raggedright\arraybackslash\p{<width>}` columns at `\textwidth`. Rows: informed consent, staging/KRAS, CA 19-9, ECOG, labs, Phase 0 sim sign-off, USL + safety matrix, surgery, intra-op telemetry, ISGPS fistula grading, daraxonrasib restart advisory, AE/SAE review, imaging (RECIST), PFS/OS. Columns: Screening (Day -28..-1), Baseline (Day -1), Surgery (Day 0), Acute (Day 1-7), Day 30, Day 90, Long-term (q12wk). Source the visit map from *mermaid/fig-14-schedule-of-activities.md* and timings from *research/ChatGPT-5.5-Thinking-Extended-19Jun26.md*.

3 Introduction

3.1 Study Rationale

Pancreatic ductal adenocarcinoma is the third leading cause of cancer death in the United States with five-year overall survival below 13 percent, and the Whipple procedure is its most technically demanding curative-intent operation. This protocol pairs a precise, auditable, on-premises LLM-directed robotic Whipple with the RAS(ON) multi-selective inhibitor daraxonrasib.

[DRAFTING INSTRUCTION] *Develop the rationale around the three counterfactual scenarios in which withholding the LLM + robot + medicine combination shortens PFS and OS; render mermaid/fig-19-counterfactual-scenarios.md as a TikZ figure. Politely raise the framing questions: how did we ever rely on humans alone for this treatment; why did we fully trust a prior trial system with so many opportunities for compounding human error; why would human peer review continue for increasingly complex and voluminous AI outputs. Source the clinical cascade from inputs/2030-pdac-1min-final-paper/sections/{introduction,discussion}.tex.*

3.2 Background

[DRAFTING INSTRUCTION] *Summarize: (a) PDAC epidemiology and the Whipple fistula-sepsis cascade (2030-pdac introduction); (b) daraxonrasib mechanism, FDA Breakthrough Therapy designation, RASolute 302 / RASolve 301 (cite the five DARAXONRASIB entries, brief explanation of each); (c) the current state of AI in trials - prior AI was prediction-only or narrow, 0 generative LLM devices approved, 1,016 to 1,450+ narrow AI/ML authorizations - from research/{ChatGPT-5.5-Thinking-Extended-18Jun26,Gemini-3.1-Pro-18Jun26}.md; (d) the author's established LLM trust, Aug 2024-Jun 2026, from inputs/author_works.bib, rendered via mermaid/fig-23-author-trust-timeline.md, with brief descriptions only.*

[DRAFTING INSTRUCTION] *Add a Background subsection "Physical AI concerns addressed by this protocol" covering all eight items (limitations; patient safety; loss of human workers; single-source software; proprietary-model dependency; non-domestic open-source LLMs; overly complex workflows; black boxes), each paired with its mitigation; render mermaid/fig-20-physical-ai-concerns.md as a TikZ figure and as a full-width table.*

3.3 Risk/Benefit Assessment

3.3.1 Known Potential Risks

[DRAFTING INSTRUCTION] *Enumerate drug risks (daraxonrasib toxicity), device/robotic risks (malfunction, AI advisory error, cyber-physical failure, vascular injury), and operative risks (fistula, hemorrhage, conversion). Source robotic risk categories from inputs/21cfr312_adapt/01_preamble_scope_definitions.tex.*

3.3.2 Known Potential Benefits

[DRAFTING INSTRUCTION] *Precise in-window R0 resection, layered hard safety limits with full audit trail, optimized drug-restart timing; for a low-survival population.*

3.3.3 Assessment of Potential Risks and Benefits

[DRAFTING INSTRUCTION] *Render mermaid/fig-24-risk-benefit-terminal.md as a TikZ figure; conclude the benefit-risk balance is favorable for advanced patients, consistent with the expanded-access ethic (21 CFR 312.84; research/ChatGPT-5.5-Thinking-Extended-19Jun26.md).*

4 Objectives and Endpoints

The trial pairs each objective with a measurable endpoint and a justification, in the standard three-column form.

[DRAFTING INSTRUCTION] *Build the three-column Objectives / Endpoints / Justification table as a full-width table (one >\raggedright\arraybackslashp{<width>} column per field at \textwidth). Render mermaid/fig-13-objectives-endpoints.md as a companion TikZ figure. Populate from nih-protocol/02_objectives_and_endpoints.md and the device-feasibility endpoints in research/ChatGPT-5.5-Thinking-Extended-19Jun26.md.*

4.1 Primary Objectives and Endpoints

[DRAFTING INSTRUCTION] *Primary safety: incidence of device-related or procedure-related serious adverse events through 30 days after the procedure. Primary dose-finding: MTD / RP2D of perioperative daraxonrasib by 3+3. Primary technical-feasibility: proportion of procedures completing all prespecified assigned tasks without unplanned conversion caused by device malfunction or unsafe behavior.*

4.2 Secondary Objectives and Endpoints

[DRAFTING INSTRUCTION] *R0 resection rate; ISGPS grade B/C postoperative pancreatic fistula rate; Clavien-Dindo grade III+ complication rate; 30-day and 90-day mortality. Provide the human comparator values from the Dutch 2025 cohort (conversion 10.1%, CD III+ 41.3%, ISGPS B/C 24.4%, 30-day mortality 3.9%) from `inputs/2030-pdac-1min-final-paper/sections/results.tex`.*

4.3 Exploratory Objectives and Endpoints

[DRAFTING INSTRUCTION] *Progression-free survival and overall survival to 24 months; LLM advisory concordance with the safety gate; sim-to-real gap (trajectory and force). Mark all exploratory analyses as non-confirmatory.*

5 Study Design

5.1 Overall Design

A prospective, open-label, single-arm, first-in-human study combining a Phase 1 3+3 dose-finding evaluation of perioperative daraxonrasib with an early feasibility evaluation of the on-premises LLM-directed eight-arm robotic Whipple (significant-risk device, Class II collaborative).

[DRAFTING INSTRUCTION] *State the hypothesis, the combined IND/IDE phase designation, the single-arm rationale, n up to 18, and the staged-autonomy frame; render `mermaid/fig-18-staged-autonomy.md`. Source the Class II / continuous oversight rules from `inputs/21cfr312_adapt/05_clinical_holds_appendices_closing.tex`.*

5.2 Scientific Rationale for Study Design

[DRAFTING INSTRUCTION] *Justify single-arm first-in-human (vs randomized) by the device-feasibility precedent and the vulnerable population; cite `research/ChatGPT-5.5-Thinking-Extended-19Jun26.md`.*

5.3 Justification for Dose

[DRAFTING INSTRUCTION] *Justify daraxonrasib starting dose and 3+3 escalation in exact doses (not percentages) anchored to RASolute 302; render `mermaid/fig-10-dose-escalation.md`. Justify the device “dose” analogue: USL ≥ 7.0 surgical readiness, Phase 0 ≥ 1000 simulated procedures across ≥ 2 frameworks, sim-to-real $< 2\text{mm}$ / $< 0.5\text{N}$, from `inputs/21cfr312_adapt/02_ind_content_phases.tex`.*

5.4 End of Study Definition

[DRAFTING INSTRUCTION] *Adapt the NIH End-of-Study language: a participant completes the study at the last visit / procedure in the SoA; the end of study is the last participant’s 24-month OS assessment. Source from `nih-protocol/03_study_design_and_study_population.md`.*

6 Study Population

[DRAFTING INSTRUCTION] *Render `mermaid/fig-02-consort-flow.md` as a TikZ figure and state the target enrollment (n approx. 36 screened, 18 treated) consistent with sec-09 sample size.*

6.1 Inclusion Criteria

[DRAFTING INSTRUCTION] All must be met. Adapt the NIH inclusion list (*nih-protocol/03_study_design_and_study_population.md*) to this trial: signed informed consent including the Physical AI opt-out; age ≥ 18 ; histologically confirmed resectable or borderline-resectable PDAC; KRAS G12 mutation; ECOG 0-1; adequate organ/marrow function; CA 19-9 measured; daraxonrasib-eligible under RASolute 302 broad pan-KRAS criteria; anatomy compatible with the eight-arm system. Use the synthetic exemplar PAT-PDAC-0001 (68 y, 3.4 cm head-of-pancreas tumor abutting SMV at 75 deg, KRAS G12D, CA 19-9 412 U/mL, ECOG 1, post mFOLFIRINOX x4) from *inputs/2030-pdac-1min-final-paper/sections/introduction.tex*.

6.2 Exclusion Criteria

[DRAFTING INSTRUCTION] Any one excludes. Adapt: distant metastatic disease; vascular involvement precluding R0; prior pancreatic surgery; uncontrolled comorbidity; pregnancy/lactation; contraindication to daraxonrasib; inability to consent. Include the equitable-selection statement; note that limited English proficiency is not an exclusion criterion.

6.3 Lifestyle Considerations

[DRAFTING INSTRUCTION] Perioperative diet, alcohol, and tobacco restrictions; applicable or mark not applicable.

6.4 Screen Failures

[DRAFTING INSTRUCTION] Adapt the CONSORT minimal screen-failure data set (demography, screen-failure detail, eligibility, any SAE); rescreened participants retain the same participant number.

6.5 Strategies for Recruitment and Retention

[DRAFTING INSTRUCTION] Single-site academic recruitment; accrual rate; retention through the 24-month OS window; vulnerable-population safeguards (OHRP); compensation amount, form, timing, recipient.

7 Study Intervention

7.1 Study Intervention Administration

7.1.1 Study Intervention Description

[DRAFTING INSTRUCTION] Describe both interventions. Drug: daraxonrasib (RMC-6236), RAS(ON) multi-selective inhibitor; oral, perioperative. Device: the eight-arm LLM-directed robotic Whipple platform - 56 DOF, 640 sensor channels, 0.05 mm RMS at 1200 mm/s, 3 N per-arm and 18 N cumulative force caps - from *inputs/2030-pdac-1min-final-paper/sections/methods.tex*. Render *mermaid/fig-05-platform-architecture.md* and *mermaid/fig-04-llm-advisory-loop.md* as TikZ figures, and include the per-arm tool-assignment table and the ten-channel sensor table as full-width tables. Include the 21 CFR 312.23(g) Physical AI System Description fields from *inputs/21cfr312_adapt/02_ind_content_phases.tex*.

7.1.2 Dosing and Administration

[DRAFTING INSTRUCTION] Daraxonrasib 3+3 dose-escalation in exact doses with 28-day DLT window (render *mermaid/fig-10-dose-escalation.md*); the perioperative pause-and-restart advisory (render *mermaid/fig-09-daraxonrasib-advisory.md*; restart T+7/T+14/T+21 keyed to ISGPS fistula grade and the 0.5 ng/mL serum trough, 29/3/0 of 32). Device “administration”: the eight operative phases (render *mermaid/fig-06-intraoperative-timeline.md*) and the three anastomoses with ring-tension bands (render *mermaid/fig-11-three-anastomoses.md*).

7.2 Preparation, Handling, Storage, and Accountability

[DRAFTING INSTRUCTION] *Four subsections (acquisition/accountability; formulation/labeling; storage/stability; preparation) for the drug, plus device maintenance and calibration per 21 CFR 312.23(g)(xi).*

7.3 Measures to Minimize Bias: Randomization and Blinding

[DRAFTING INSTRUCTION] *Open-label single-arm; describe outcome-assessor masking for R0 pathology and independent adjudication; cite ICH E9 and nih-protocol/04_study_intervention.md.*

7.4 Study Intervention Compliance

[DRAFTING INSTRUCTION] *Daraxonrasib adherence via drug log and serum assay; device compliance via the pre-procedure safety matrix and the hash-chained audit trail (render mermaid/fig-21-audit-trail-dataflow.md).*

7.5 Concomitant Therapy

[DRAFTING INSTRUCTION] *Permitted and prohibited concomitant medications; perioperative supportive care; record in the CRF.*

7.5.1 Rescue Medicine

[DRAFTING INSTRUCTION] *Rescue for surgical complications and drug toxicity; list and administration instructions.*

8 Intervention and Participant Discontinuation/Withdrawal

8.1 Discontinuation of Study Intervention

[DRAFTING INSTRUCTION] *Define drug-stop and device-stop rules. Device stopping rules include intra-operative conversion to manual/open, repeated force-limit excursions, no-fly-zone violations, sim-to-real divergence beyond twice tolerance ($> 4\text{mm}$ or $> 1.0\text{N}$), and any hard-stop E-stop event; source from inputs/21cfr312_adapt/{02_ind_content_phases, 03_protocol_amendments_reporting}.tex. Drug stopping rules include DLTs and grade C fistula.*

8.2 Participant Discontinuation/Withdrawal from the Study

[DRAFTING INSTRUCTION] *Voluntary withdrawal at any time without penalty; investigator-initiated withdrawal; data retained to the point of withdrawal; relate to the analysis populations in sec-09.*

8.3 Lost to Follow-Up

[DRAFTING INSTRUCTION] *Define lost-to-follow-up procedures across the 24-month OS window and the missing-data handling that links to sec-09.4.*

9 Study Assessments and Procedures

9.1 Efficacy Assessments

[DRAFTING INSTRUCTION] *R0 margin by pathology; RECIST imaging; PFS/OS; anastomosis quality (PJ/HJ/GJ) with the ring-tension bands from mermaid/fig-11-three-anastomoses.md; ISGPS fistula grade from inputs/2030-pdac-1min-final-paper/sections/methods.tex.*

9.2 Safety and Other Assessments

[DRAFTING INSTRUCTION] *Vital signs, labs, imaging; device telemetry (force profiles, trajectory accuracy, E-stop events, vessel-zone verdicts) from mermaid/fig-07-vascular-safety-gate.md and fig-08-estop-architecture.md; the pre-procedure safety matrix.*

9.3 Adverse Events and Serious Adverse Events

[DRAFTING INSTRUCTION] *Render mermaid/fig-15-ae-reporting.md as a TikZ figure.*

9.3.1 Definition of Adverse Events (AE)

[DRAFTING INSTRUCTION] *CTCAE v5 clinical AE definition plus the Physical AI AE definition (robotic malfunction, AI prediction error, sensor failure, comms loss, unauthorized access, digital-twin divergence, any E-stop) from inputs/21cfr312_adapt/03_protocol_amendments_reporting.tex.*

9.3.2 Definition of Serious Adverse Events (SAE)

[DRAFTING INSTRUCTION] *Standard SAE plus serious Physical AI AE (unplanned interruption requiring clinical intervention, uncontrolled drug release, sterile-field compromise).*

9.3.3 Classification of an Adverse Event

[DRAFTING INSTRUCTION] *Severity (CTCAE), relationship to drug and to device, expectedness.*

9.3.4 Time Period and Frequency for Assessment and Follow-Up

[DRAFTING INSTRUCTION] *Through the 30-day primary safety window and the long-term OS follow-up.*

9.3.5 Adverse Event Reporting

[DRAFTING INSTRUCTION] *Routine CRF capture.*

9.3.6 Serious Adverse Event Reporting

[DRAFTING INSTRUCTION] *7-day (fatal/life-threatening) and 15-day (serious/unexpected) IND/IDE timelines; the six Physical AI reporting triggers; audit-trail preservation from -24h to +72h around the event.*

9.3.7 Events of Special Interest

[DRAFTING INSTRUCTION] *Vascular injury, pancreatic fistula, conversion, cybersecurity incident, model-performance degradation.*

9.3.8 Reporting of Pregnancy

[DRAFTING INSTRUCTION] *Standard pregnancy reporting for daraxonrasib exposure.*

9.4 Unanticipated Problems

[DRAFTING INSTRUCTION] *Definition, reporting to the IRB and FDA, and reporting to participants; tie to the Physical AI hold grounds in `inputs/21cfr312_adapt/05_clinical_holds_appendices_closing.tex`.*

10 Statistical Considerations

10.1 Statistical Hypotheses

[DRAFTING INSTRUCTION] *State the primary safety hypothesis (the device-related serious-AE rate is below a prespecified unacceptable threshold) and the dose-finding objective (identify MTD/RP2D). Cite ICH E9 and `nih-protocol/07_statistical_considerations.md`.*

10.2 Sample Size Determination

[DRAFTING INSTRUCTION] *Justify up to $n=18$ by the 3+3 design and the early-feasibility precedent (generally fewer than 10 per device-feasibility guidance, expanded for dose finding); state the operating characteristics rather than power for a single-arm safety study; source from `research/ChatGPT-5.5-Thinking-Extended-19Jun26.md`.*

10.3 Populations for Analyses

[DRAFTING INSTRUCTION] *Define Safety (all dosed), DLT-evaluable, Per-Protocol, and modified ITT populations; render `mermaid/fig-22-analysis-populations.md`.*

10.4 Statistical Analyses

10.4.1 General Approach

[DRAFTING INSTRUCTION] *Descriptive statistics with 95% confidence intervals; exact methods for proportions given the small n .*

10.4.2 Analysis of the Primary Endpoint(s)

[DRAFTING INSTRUCTION] *30-day device/procedure-related serious-AE incidence with exact CI; MTD/RP2D determination; task-completion proportion.*

10.4.3 Analysis of the Secondary Endpoint(s)

[DRAFTING INSTRUCTION] *R0 rate, ISGPS B/C fistula, Clavien-Dindo III+, mortality, each with exact CIs and the Dutch 2025 comparator.*

10.4.4 Safety Analyses

[DRAFTING INSTRUCTION] *MedDRA-coded AEs by system organ class; device telemetry summaries; Physical AI AE tabulation.*

10.4.5 Baseline Descriptive Statistics

[DRAFTING INSTRUCTION] *Demographics, staging, KRAS, CA 19-9, ECOG.*

10.4.6 Planned Interim Analyses

[DRAFTING INSTRUCTION] Cohort-level DSMB safety review with prespecified halt rules; render the stopping logic from `mermaid/fig-22-analysis-populations.md`.

10.4.7 Sub-Group Analyses

[DRAFTING INSTRUCTION] By resectable vs borderline-resectable; by dose level.

10.4.8 Tabulation of Individual Participant Data

[DRAFTING INSTRUCTION] Listings by participant and timepoint.

10.4.9 Exploratory Analyses

[DRAFTING INSTRUCTION] PFS/OS, LLM advisory concordance, sim-to-real gap; non-confirmatory.

11 Regulatory, Ethical, and Oversight Considerations

[DRAFTING INSTRUCTION] Render `mermaid/fig-16-governance-oversight.md` as a TikZ figure to open the governance section.

11.1 Informed Consent Process

[DRAFTING INSTRUCTION] Render `mermaid/fig-17-informed-consent.md`; describe the consent and the Physical AI opt-out (21 CFR 312.60(f)), assent where applicable, and documentation. Source from `inputs/21cfr312_adapt/05_clinical_holds_appendices_closing.tex` and `nih-protocol/08_supporting_documentation_regulatory_oversight.md`.

11.2 Study Discontinuation and Closure

[DRAFTING INSTRUCTION] FDA/IRB-driven closure; Physical AI clinical-hold grounds and the decommissioning/data-preservation procedures (21 CFR 312.38, 312.42).

11.3 Confidentiality and Privacy

[DRAFTING INSTRUCTION] HIPAA Safe Harbor de-identification (18 identifiers), deny-by-default authorization, the hash-chained audit trail (render `mermaid/fig-21-audit-trail-dataflow.md`); 21 CFR part 11.

11.4 Future Use of Stored Specimens and Data

[DRAFTING INSTRUCTION] Telemetry pyramid retention (render `mermaid/fig-25-sensor-data-pyramid.md`); L0 raw on Zenodo, decompressible on FDA request per 21 CFR 312.57.

11.5 Key Roles and Study Governance

[DRAFTING INSTRUCTION] Sponsor-Investigator, site PI, operator and backup operator, CRO; duties from `inputs/21cfr312_adapt/04_annual_reports_withdrawal.tex`.

11.6 Safety Oversight

[DRAFTING INSTRUCTION] DSMB, Independent Safety Monitor, and the Physical AI Safety Review Committee on a ≤ 90 -day cadence.

11.7 Clinical Monitoring

[DRAFTING INSTRUCTION] *Robotics- and AI-competent monitors; source-data verification of telemetry and audit trails.*

11.8 Quality Assurance and Quality Control

[DRAFTING INSTRUCTION] *Pre-procedure safety matrix verification; USL reassessment; Phase 0 re-validation after any update.*

11.9 Data Handling and Record Keeping

[DRAFTING INSTRUCTION] *The seven Physical AI record types and retention per 21 CFR 312.57.*

11.10 Protocol Deviations

[DRAFTING INSTRUCTION] *Definition, documentation, and reporting; immediate-hazard modifications per 21 CFR 312.30(h).*

11.11 Publication and Data Sharing Policy

[DRAFTING INSTRUCTION] *GitHub + Zenodo dual deposition; CC BY 4.0; TRIPOD+AI and CREMLS reporting.*

11.12 Conflict of Interest Policy

[DRAFTING INSTRUCTION] *Author and sponsor disclosures; 21 CFR part 54.*

12 Additional Considerations, Abbreviations, and Amendment History

12.1 Additional Considerations

[DRAFTING INSTRUCTION] *Cross-jurisdictional SaMD tracks (EU MDR, NMPA, PMDA, MHRA, Health Canada, TGA) and the H. R. 9510 VVUQ legislative/financial alignment from `inputs/auto-bill-02`; the ten-gate assurance suite (render `mermaid/fig-12-vvuq-ten-gate.md`).*

12.2 Abbreviations

[DRAFTING INSTRUCTION] *Build the glossary as a full-width two-column table: AE, SAE, CTCAE, DLT, DSMB, ECOG, FDA, GCP, HJ, ICH, IDE, IND, ISGPS, ISM, ITT, KRAS, LLM, MCP, MTD, OS, PDAC, PFS, PJ, GJ, PV, RP2D, SMA, SMV, USL, VVUQ. Resolve every acronym used across sections.*

12.3 Protocol Amendment History

[DRAFTING INSTRUCTION] *Version-tracking table; v4.0.0 is the initial protocol version dated June 20, 2026.*

13 References and Back Matter

[DRAFTING INSTRUCTION] *Render the References via `ieeetr` from `references.bib`: the five DARAX-ONRASIB entries; the relevant FDA guidance and 21 CFR parts 11/50/812 from the `research/` markdowns; the IEC/ASME/ NASA/IEEE standards; the NIH protocol template; and only the directly necessary author works. Cite each main document with a brief explanation.*

Acknowledgments

This protocol was assembled with Claude Code (Opus 4.8, 1M context). The author thanks the open-source robotics, simulation, and clinical-standards communities.

Author and ORCID

[DRAFTING INSTRUCTION] *Kevin Kawchak, CEO ChemicalQDevice, ORCID 0009-0007-5457-8667; render the ORCID iD mark via \orcidicon and the ORCID URL.*

Ethical Disclosures

[DRAFTING INSTRUCTION] *No real patients, PHI, or IRB approval; the exemplar PAT-PDAC-0001 is synthetic; all clinical figures derive from the author's simulation sources.*

Data Availability

[DRAFTING INSTRUCTION] *GitHub plus Zenodo; deterministic seed; no PII; DOI 10.5281/zenodo.xxxxxxx.*

Cite This Article

[DRAFTING INSTRUCTION] *Kawchak K. A Phase 1, First-in-Human, Combined IND/IDE Clinical Trial Protocol ... Zenodo. 2026; 10.5281/zenodo.xxxxxxx.*

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